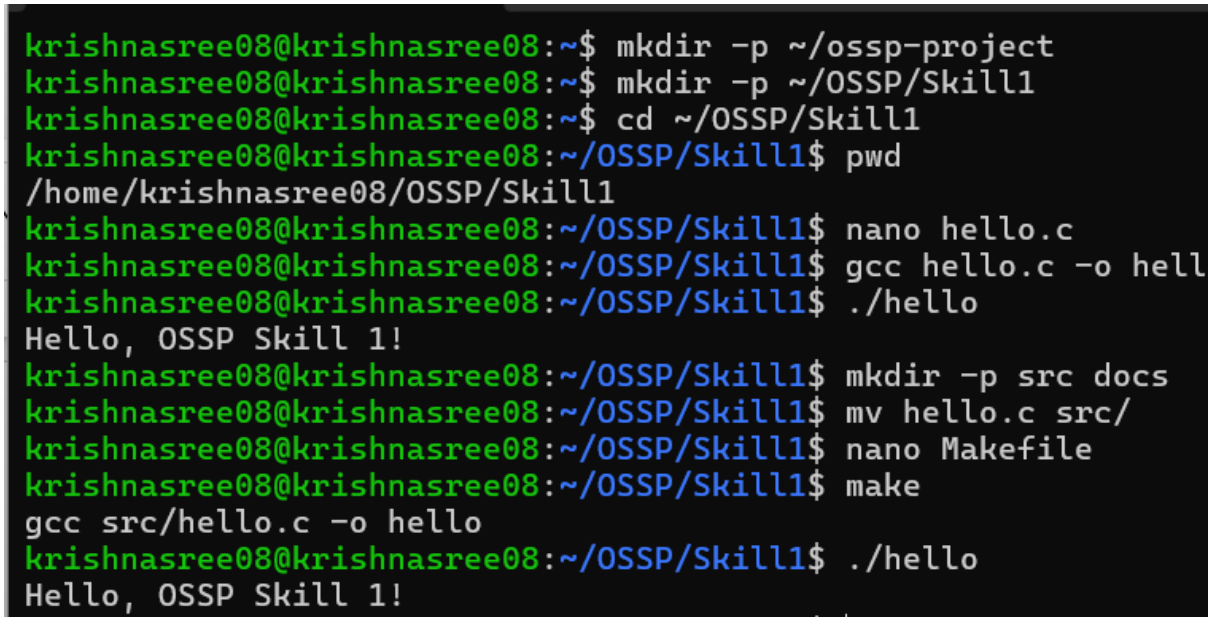
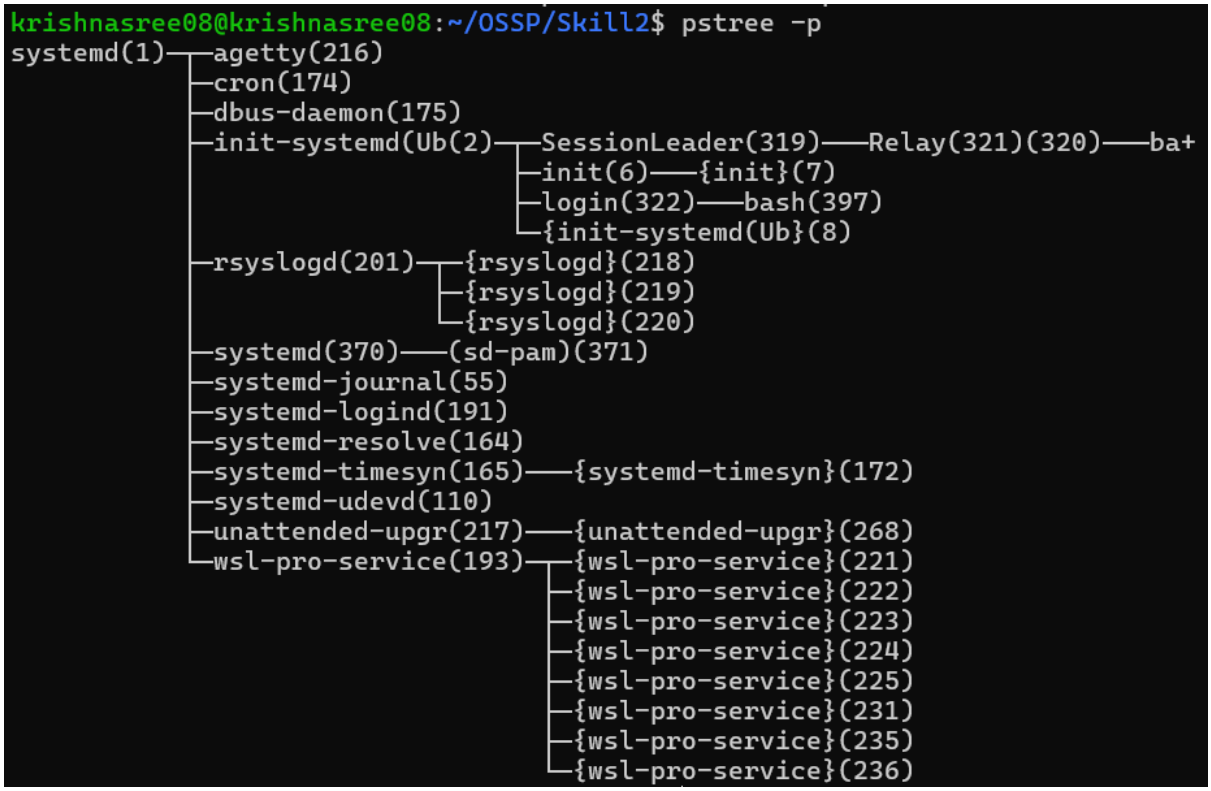


Skill 1 – Result: Linux development tools were configured, a project structure was created, and a C program was successfully compiled and executed using GCC and Makefile.

SKILL 2

```
krishnasree08@krishnasree08:~/OSSP/Skill2$ nano fork.c
krishnasree08@krishnasree08:~/OSSP/Skill2$ gcc fork.c -o fork
krishnasree08@krishnasree08:~/OSSP/Skill2$ ./fork
Child Process: PID = 542
Parent Process: PID = 541
krishnasree08@krishnasree08:~/OSSP/Skill2$ nano exec.c
krishnasree08@krishnasree08:~/OSSP/Skill2$ gcc exec.c -o exec
krishnasree08@krishnasree08:~/OSSP/Skill2$ ./exec
Before exec()
total 40
-rwxr-xr-x 1 krishnasree08 krishnasree08 16128 Oct 26 13:52 exec
-rw-r--r-- 1 krishnasree08 krishnasree08   194 Oct 26 13:52 exec.c
-rwxr-xr-x 1 krishnasree08 krishnasree08 16128 Oct 26 13:52 fork
-rw-r--r-- 1 krishnasree08 krishnasree08   356 Oct 26 13:52 fork.c
krishnasree08@krishnasree08:~/OSSP/Skill2$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 13:48 ?        00:00:01 /sbin/init
root           2       1  0 13:48 hvc0     00:00:00 /init
root           6       2  0 13:48 hvc0     00:00:00 plan9 --control-socket 7
root          55       1  0 13:48 ?        00:00:00 /usr/lib/systemd/systemd
root         110       1  0 13:48 ?        00:00:00 /usr/lib/systemd/systemd
systemd+     164       1  0 13:48 ?        00:00:00 /usr/lib/systemd/systemd
systemd+     165       1  0 13:48 ?        00:00:00 /usr/lib/systemd/systemd
root         174       1  0 13:48 ?        00:00:00 /usr/sbin/cron -f -P
message+     175       1  0 13:48 ?        00:00:00 @dbus-daemon --system --
root         191       1  0 13:48 ?        00:00:00 /usr/lib/systemd/systemd
root         193       1  0 13:48 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog       201       1  0 13:48 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root         216       1  0 13:48 tty1     00:00:00 /sbin/agetty -o -p -- \u
root         217       1  0 13:48 ?        00:00:00 /usr/bin/python3 /usr/sh
root         319       2  0 13:48 ?        00:00:00 /init
root         320     319  0 13:48 ?        00:00:00 /init
krishna+     321     320  0 13:48 pts/0    00:00:00 -bash
root         322       2  0 13:48 pts/1    00:00:00 /bin/login -f
krishna+     370       1  0 13:48 ?        00:00:00 /usr/lib/systemd/systemd
krishna+     371     370  0 13:48 ?        00:00:00 (sd-pam)
krishna+     397     322  0 13:48 pts/1    00:00:00 -bash
krishna+     550     321 50 13:53 pts/0    00:00:00 ps -ef
```



Skill 2 – Result: Process creation using fork(), program execution using exec(), parent-child relationships, process-tree inspection, and system call tracing were studied and demonstrated.